

MATHLETICS

Trigonometry

Teacher Book - Series J-1

$\sin \theta$ $\cos \theta$
 $\tan \theta$



Mathletics
Instant
Workbooks



Trigonometry

Topic Test

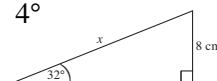
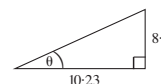
PART A

Time allowed: 15 minutes

Total marks = 15

Marks

- 1 Use your calculator to find $\cos 48^\circ$ correct to two decimal places.
 (A) 0.74 (B) 1.11 (C) 0.67 (D) none of these
- 2 Evaluate $25 \tan 63^\circ$ correct to two decimal places.
 (A) 1.96 (B) 49.07 (C) 29.38 (D) 22.28
- 3 Find the value of $\frac{\cos 32^\circ}{43.27}$ correct to two decimal places.
 (A) 0.01 (B) 0.02 (C) 0.03 (D) 0.0196
- 4 If $\sin \theta = \frac{5}{9}$, calculate the size of angle θ to the nearest degree.
 (A) 31° (B) 32° (C) 33° (D) 34°
- 5 The hypotenuse of a right-angled triangle is 41 cm. If one side is 40 cm, the third side is
 (A) 1 cm (B) 9 cm (C) 10 cm (D) 81 cm
- 6 If $\cos \theta = \frac{1}{2}$, find the size of angle θ .
 (A) 30° (B) 45° (C) 60° (D) 72°
- 7 28.65° equals
 (A) $29^\circ 5'$ (B) $28^\circ 39'$ (C) $29^\circ 39'$ (D) $28^\circ 5'$
- 8 The three sides of a right-angled triangle measure 15 cm, 17 cm and 8 cm. The length of the hypotenuse is
 (A) 8 cm (B) 15 cm (C) 17 cm (D) 25 cm
- 9 Find the size of angle θ to the nearest degree.
 (A) 40° (B) 41° (C) 42° (D) 58°
- 10 In a $\triangle ABC$, the angle B is 90° , AB is 8 m and AC is 10 m. Find the size of angle A correct to the nearest degree.
 (A) 36° (B) 39° (C) 53° (D) 37°
- 11 A road rises uniformly 30.6 m for every 600 m along the road. Find the angle of elevation of this road correct to the nearest degree.
 (A) 1° (B) 2° (C) 3° (D) 4°
- 12 Find the hypotenuse of this triangle in centimetres correct to 1 decimal place.
 (A) 9 cm (B) 15.1 cm (C) 12.8 cm (D) none of these
- 13 Use your calculator to find $7.9 \cos 63^\circ$ correct to three significant figures.
 (A) 3.58 (B) 3.59 (C) 7.03 (D) 7.04
- 14 Evaluate $\frac{\sin 54^\circ}{28.65}$ correct to two decimal places.
 (A) 0.02 (B) 0.03 (C) 0.04 (D) 0.05
- 15 Find the size of the acute angle θ to the nearest degree if $\tan \theta = \frac{12.5}{19.34}$
 (A) 40° (B) 32° (C) 33° (D) none of these



Total marks achieved for PART A

15

Trigonometry

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PART B

Time allowed: 15 minutes

Total marks = 15

Marks

Question 1

a Find the value of each expression correct to two decimal places.

i $\frac{\cos 72^\circ}{8.93} =$ _____

ii $\frac{72.54}{\tan 68^\circ} =$ _____

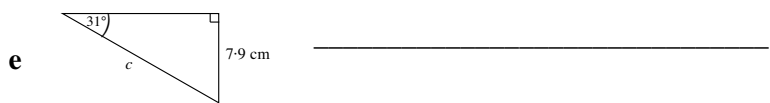
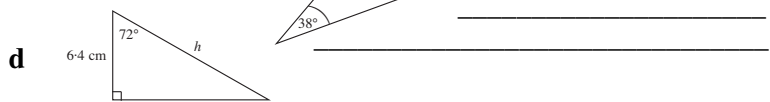
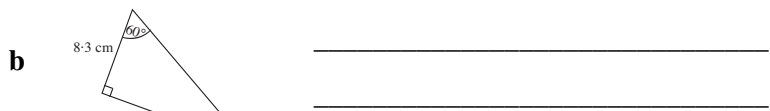
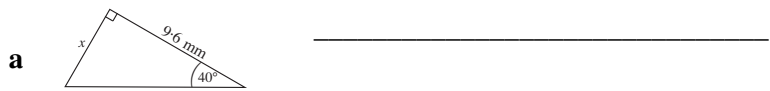
iii $\frac{34.20}{\sin 56^\circ} =$ _____

b Find acute angle A to the nearest degree.

i $\sin A = 0.6835$ _____

ii $\tan A = 1.4862$ _____

Question 2 Find the value of the pronumeral in each triangle correct to two decimal places.



Question 3 The diagonal of a square is 12.5 cm long.

a Find the length of one side correct to the nearest mm.

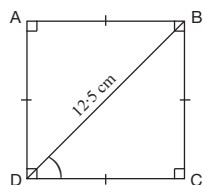
b Find the size of $\angle BDC$.

Find the following correct to three decimal places.

c $\sin \angle BDC$

d $\cos \angle DBC$

e $\tan \angle ABD$



Total marks achieved for PART B

15

Answers – Trigonometry

Page 1 1 a opp, adj, hyp b hyp, adj, opp c adj, hyp, opp 2 a a = opp, b = adj, c = hyp b d = opp, e = adj, f = hyp c g = opp, i = adj, h = hyp d k = opp, l = adj, j = hyp e n = opp, m = adj, o = hyp f r = opp, q = adj, p = hyp 3 a AB b DF c GI d JK e MN f PQ

Page 2 1 a $\frac{40}{41}$ b $\frac{5}{13}$ c $\frac{8}{17}$ 2 a $\sin \theta = \frac{1}{\sqrt{17}}$, $\sin \alpha = \frac{4}{\sqrt{17}}$ b $\sin \theta = \frac{a}{b}$, $\sin \alpha = \frac{2}{b}$ c $\sin \theta = \frac{m}{n}$, $\sin \alpha = \frac{6}{n}$ 3 a BC = 5, $\sin \theta = \frac{5}{13}$

b EF = 4, $\sin \theta = \frac{4}{5}$ c PQ = 37, $\sin \theta = \frac{12}{37}$ 4 a $\sin A = \frac{3}{5}$, $\sin B = \frac{4}{5}$ b $\sin P = \frac{5}{13}$, $\sin Q = \frac{12}{13}$ c $\sin D = \frac{9}{41}$, $\sin E = \frac{40}{41}$ 5 a $\sin A$, $\sin B$ b $\sin D$, $\sin E$ c $\sin P$, $\sin Q$

Page 3 1 a $\frac{4}{5}$ b $\frac{15}{17}$ c $\frac{3}{\sqrt{58}}$ 2 a $\cos \theta = \frac{8}{b}$, $\cos \alpha = \frac{a}{b}$ b $\cos \theta = \frac{p}{q}$, $\cos \alpha = \frac{5}{q}$ c $\cos \theta = \frac{x}{y}$, $\cos \alpha = \frac{4}{y}$ 3 a AB = $\sqrt{58}$, $\cos \theta = \frac{7}{\sqrt{58}}$

b DE = $\sqrt{29}$, $\cos \theta = \frac{2}{\sqrt{29}}$ c QR = $\sqrt{65}$, $\cos \theta = \frac{\sqrt{65}}{9}$ 4 a $\cos A = \frac{3}{5}$, $\cos B = \frac{4}{5}$ b $\cos D = \frac{12}{13}$, $\cos E = \frac{5}{13}$ c $\cos P = \frac{15}{17}$, $\cos Q = \frac{8}{17}$

5 a $\cos A$, $\cos B$ b $\cos D$, $\cos E$ c $\cos P$, $\cos Q$

Page 4 1 a $\frac{1}{3}$ b $\frac{5}{2}$ c $\frac{5}{12}$ 2 a $\tan \theta = \frac{b}{a}$, $\tan \alpha = \frac{a}{b}$ b $\tan \theta = \frac{p}{q}$, $\tan \alpha = \frac{8}{p}$ c $\tan \theta = \frac{7}{a}$, $\tan \alpha = \frac{a}{7}$ 3 a AC = $\sqrt{56}$, $\tan \theta = \frac{5}{\sqrt{56}}$

b DF = 7, $\tan \theta = \frac{12}{7}$ c QR = $\sqrt{133}$, $\tan \theta = \frac{6}{\sqrt{133}}$ 4 a $\tan A = \frac{4}{3}$, $\tan B = \frac{3}{4}$ b $\tan D = \frac{9}{40}$, $\tan E = \frac{40}{9}$ c $\tan P = \frac{4}{3}$, $\tan Q = \frac{3}{4}$

5 a $\tan A$, $\tan B$ b $\tan D$, $\tan E$ c $\tan P$, $\tan Q$

Page 5 1 a 0.56 b 2.75 c 0.97 d 0.52 e 0.77 f 0.62 g 8.14 h 0.50 i 2.05 2 a 0.287 b 0.055 c 23.073 d 0.312 e 0.059 f 25.519 g 0.077 h 0.356 i 1542.844 3 a 1.66 b 1.49 c 0.814 d 4.02 e 0.428 f 0.935 g 0.475 h 3.50 i 0.440 4 a 39° b 68° c 68° d 55° e 38° f 50° g 59° h 16° i 10° 5 a 30° b 30° c 50° d 67° e 60° f 65° 6 a 35° b 67° c 35° d 30° e 42° f 40°

Page 6 1 a x = 4.2 b a = 7.6 c y = 13.1 2 a n = 28.14 b m = 9.77 c p = 9.92 3 a q = 3.35 b t = 18.36 c l = 12.16 d c = 6.27 e d = 23.90 f k = 10.01 4 1.94 m 5 5.75 cm

Page 7 1 a 16.6 cm b 15.6 cm c 16.7 cm 2 a 16.00 cm b 14.15 cm c 19.30 cm 3 a 31.4 cm b 34.7 cm c 24.4 cm d 20.2 cm e 12.0 cm f 19.5 cm 4 x = 24 cm 5 43 m

Page 8 1 a $\theta = 31^\circ$ b $\alpha = 57^\circ$ c $\beta = 76^\circ$ 2 a $\beta = 68^\circ$ b $\theta = 24^\circ$ c $\alpha = 37^\circ$ 3 a $\beta = 71^\circ$ b $\alpha = 35^\circ$ c $\theta = 60^\circ$ d $\theta = 74^\circ$ e $\alpha = 69^\circ$ f $\beta = 23^\circ$ 4 $\angle P = 51^\circ$ 5 $\angle ACD = 25^\circ$

Page 9 1 a 564 m b 81.04 m c 2° 2 a 27 m b 54° c 55°

Page 10 1 C 2 B 3 B 4 D 5 B 6 C 7 B 8 C 9 A 10 D 11 C 12 B 13 B 14 B 15 C

Page 11 1 a i 0.03 ii 29.31 iii 41.25 b i 43° ii 56° 2 a x = 8.06 cm b m = 14.38 cm c y = 9.73 cm d h = 20.71 cm e c = 15.34 cm 3 a 8.8 cm b 45° c 0.707 d 0.707 e 1.000